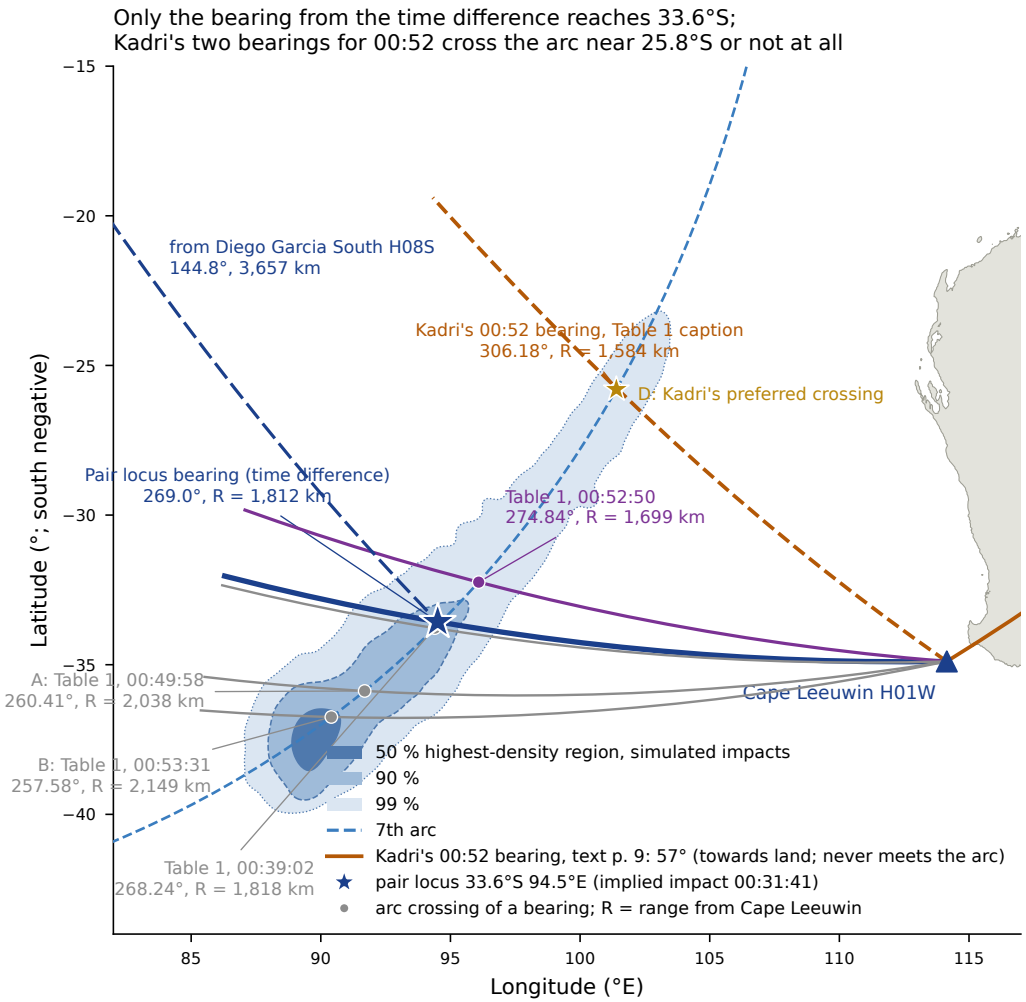


(i) Each line is a sound direction from Cape Leeuwin. The dot is where the line meets the arc, and R is the distance. The thick blue line goes to the place that the time difference of the pair gives. Kadri's two published directions for the 00:52 signal (57° and 306.18°) do not go there. The grey lines are other Table 1 directions that the earlier chart used; the 00:39:02 direction meets the arc 0.2° from the star, 13 minutes earlier. The two sounds (Cape Leeuwin 00:52:04, Diego Garcia 01:12:49) have a time difference that agrees with one point on the arc, 33.6°S 94.5°E. Pairs like this occur by chance: the test gives p 0.019 from 36 tests, which is a chance level. The Diego Garcia sound is not kept when the airgun noise is removed. The sound direction that Kadri published for 00:52 does not point to this place. The impact would need about 330 times more sound than the calibration crash gives.

(ii) Pair G: Cape Leeuwin H01W 00:52:04 UTC (DIGITISED Kadri 2024 Fig. 9 panel c; envelope matched filter M1b p 0.043; Kadri's 'text time 00:52:03', 'caption time 00:54:30') x Diego Garcia South H08S 01:12:49 (DIGITISED panel e, raw, M1b p 0.039; the airgun-suppressed residual does not keep it, p_B 0.30); lag 1,245 s inside the predicted 809-1,536 s; 19 lag-consistent pairs vs 8.8 expected, permutation p 0.019 over 36 receiver-pair tests (chance level: a smallest p this low arises about half the time). Arc locus 33.6°S 94.5°E, implied impact 00:31:41; eta for P_D(both) = 0.5 is 0.069 = 330x the F-35A anchor 2.1e-4. BEARING: the locus lies at 269.0° from H01W (1,812 km) and 144.8° from H08S (3,657 km), WGS84; Kadri gives 57° (text p. 9) or 306.18° (Table 1 caption) for the 00:52 signal; neither is within 37° of the locus bearing; nearest Table 1 row 00:52:50 at 274.84° (46 s later). Rays are WGS84 geodesics from the H01W triad centroid; R is the geodesic range to the nearest approach to the drawn arc (composer small-circle fit to run J 00:19:29 positions); for the pair, R is the range to the locus on the module's TL-grid arc. Table 1 rows from searches named/table1_verdicts.csv (bearings DIGITISED from Kadri 2024 Table 1). HPD: run J impacts (geometry/sample.npz, waypoints seeds 1-8 stride 6 + northern supplement, Kish ESS 110k), D0', 0.1° bins, Gaussian display smoothing sigma 25 km. Locus C of the earlier chart is not legible in the original and is not redrawn. Source posterior run J, 00:19 Held Out (D0'), dive probability p 0.5; PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, +10.4/+7.4 dB per octave at H01W/H08S) and C2 (coupling efficiency not calibrated; eta swept over four decades, F-35A upper anchor). Module branch hypothesis/hydroacoustics 9b27786; searches results-data/searches_run_j.